# Relativity

# Relativity 1

Contradiction:

Maxwell
speed of light $\Longleftrightarrow$ Newtonian relativity
(Galileo transformation)

Newton / Galilei:

A. $\exists$ global time + Euclidean space

B. $\exists$ inertial frames

a. free bodies move along straight lines

b. move with uniform velocity w.r.t. each-other

C. Principle of Relativity

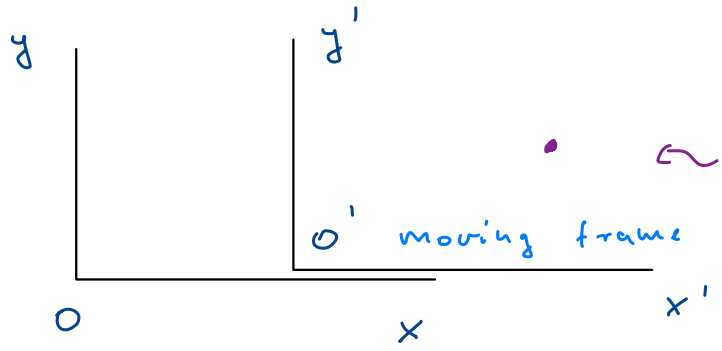
Laws of Nature are same in these
$\rightsquigarrow$ one cannot tell if one moves!
motion is relative

Einstein:

~~A.~~ B. ✓ C. ✓

light propagates with $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$
in any frame (Maxwell wins!)

# Galilean transformation:




$O'$ at $x = v \cdot t$

• ← event (clap) at time $t$ position $(x, y, z)$

$O$: $t, x, y, z$ $\Rightarrow$ $O'$:

$$t' = t$$
$$x' = x - v \cdot t$$
$$y' = y$$
$$z' = z$$

velocity: motion:

$O$: $dt \to dx$ $\Rightarrow$ $u_x = \frac{dx}{dt}$

$O'$: $x' = x - v \cdot t$ $\Rightarrow$ $dx' = dx - v \cdot dt$

$dt' = dt$

$$\Rightarrow \frac{dx'}{dt'} = \frac{dx}{dt} - v$$

$$u_x' = u_x - v$$

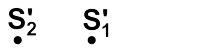
or $\underline{u}' = \underline{u} - \underline{v}$

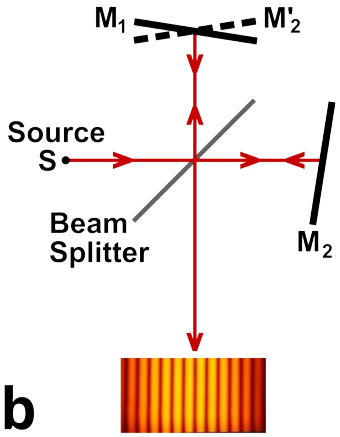
Problem: suppose light propagates in some medium (ether). If we move w.r.t this medium the its velocity should change

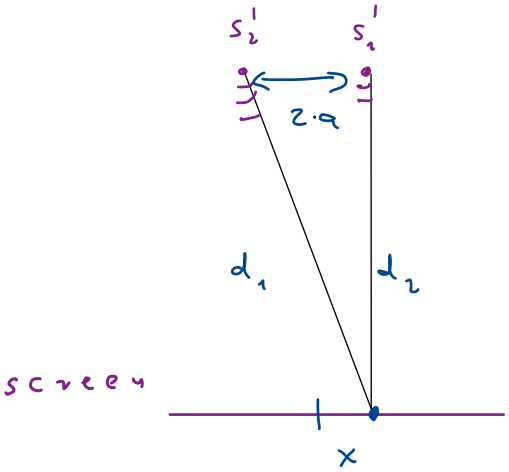
# Michelson - Morley experiment

try detect ether (or aether)

interference



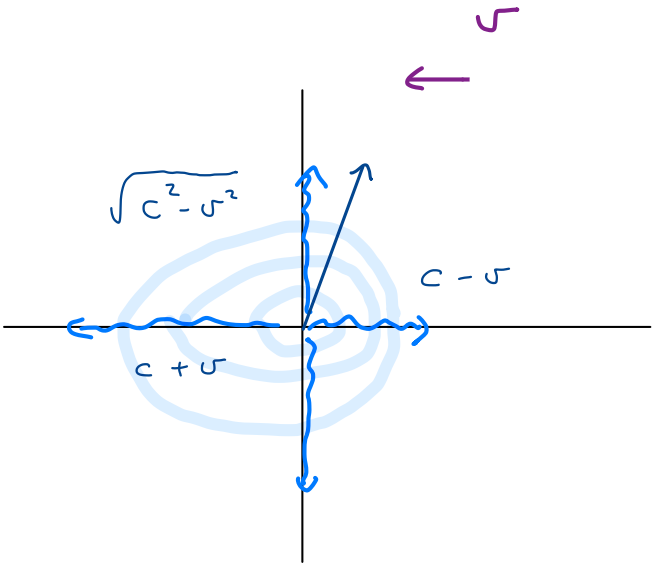




$$d_1 = \sqrt{(x-a)^2 + L^2} = L + \frac{(x-a)^2}{L}$$

$$d_2 = L + \frac{(x+a)^2}{L}$$

constructive interference $d_2 - d_1 \approx \frac{4a}{L} \cdot x = n \cdot \lambda$

"Ether wind"




if ∃ ether ⇒

$$t_{\rightarrow} = \frac{L}{c-v} \qquad t_{\leftarrow} = \frac{L}{c+v} \qquad t_{\uparrow} = \frac{L}{\sqrt{c^2 - v^2}}$$

interference condition:

$$t_{\rightarrow} + t_{\leftarrow} \equiv t_{\uparrow} + t_{\downarrow} + n \cdot T$$

assume $L_x = L_y = L$ : $\qquad f = \frac{1}{T}$

$$n \cdot T = \frac{L}{c-v} + \frac{L}{c+v} - \frac{2L}{\sqrt{c^2 - v^2}} \approx \frac{L}{c} \frac{v^2}{c^2}$$

$$n \lambda_0 = L \cdot \frac{v^2}{c^2}$$

rotation by $90^\circ$ $\Rightarrow$ $t_{\leftrightarrow} \Leftrightarrow t_{\updownarrow}$

$\Rightarrow$ $\qquad n \lambda_0 = - L \frac{v^2}{c^2}$ ↯

no interference

if $2 \cdot L \frac{v^2}{c^2} \sim \lambda_0 \cdot 0.1$

take $L = 10\,m$, $\lambda_0 = 500\,nm$, $v_{earth} \approx 30 \frac{km}{s}$

$\Rightarrow$ one should see a motion of fringes

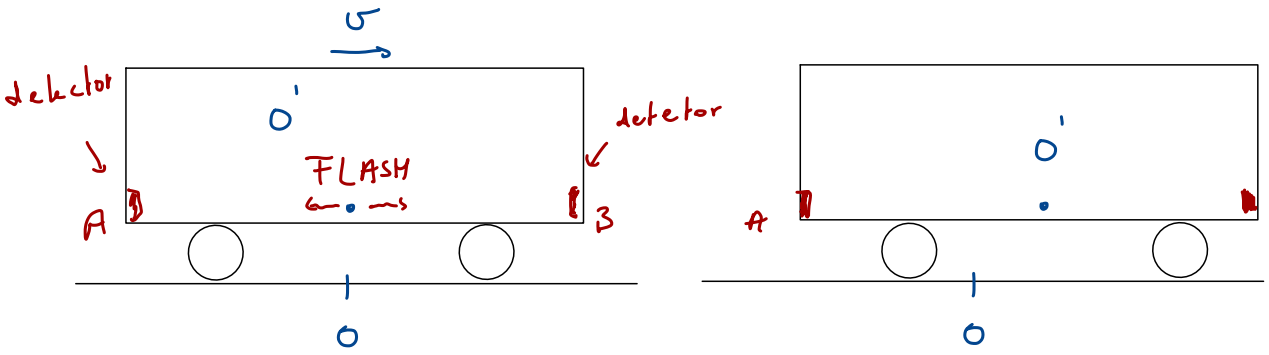
interference pattern should shift

NO SHIFT ↯ ↯

Einstein: Ether is not measurable, light propagates with same velocity for any observer

# CONSEQUENCES:

- No simultaneity

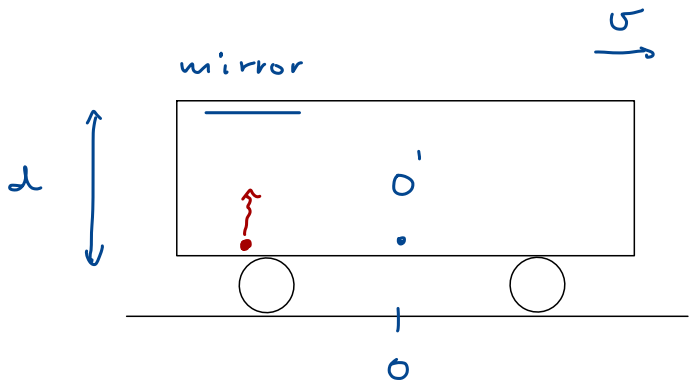



for O': detection simultaneous since light propagates in both directions with c

for O: detection is NOT simultaneous

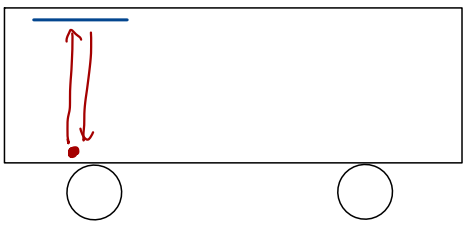
light propagates in both directions with c, but carbox moves to the right, detection at A preceeds detection at B

# • Time dilation




$T_{reflection} = ?$

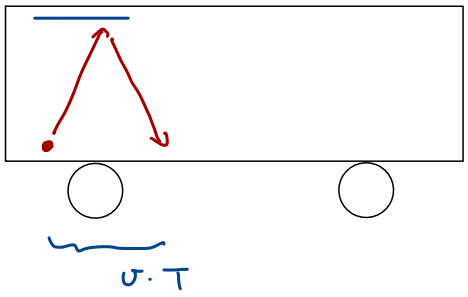
O'



$$T' = T_0 = 2\frac{d}{c}$$

proper time

O




$$2\sqrt{d^2 + \frac{v^2 T^2}{4}} = c \cdot T$$

$$\Rightarrow \quad T = \frac{2d}{\sqrt{c^2 - v^2}} = \frac{T'}{\sqrt{1 - v^2/c^2}}$$

time dilation

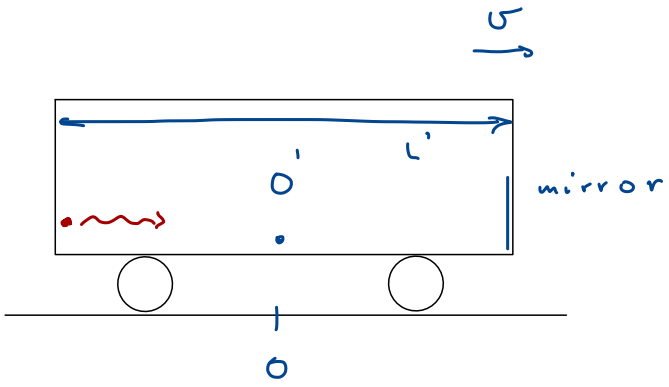
$$T = T_0 \frac{1}{\sqrt{1 - v^2/c^2}}$$

measured time

proper time (co-moving)

many experiments verified this !
(Muon lifetimes, atomic clocks on airplanes, etc. )

# Length contraction (Lorentz contraction)




O': $T' = 2L'/c$ $\quad L' = L_0$ is proper length

O: $T = ?$ forward $\quad t_1 c = t_1 v + L \Rightarrow t_1 = \frac{L}{c-v}$

backward $\quad t_2 c = L - t_2 \cdot v \Rightarrow t_2 = \frac{L}{c+v}$

$$\Rightarrow \quad T = t_1 + t_2 = \frac{2c \cdot L}{c^2 - v^2} = \frac{2L}{c} \cdot \frac{1}{1 - v^2/c^2}$$

however $\quad T = \frac{T'}{(1 - v^2/c^2)^{1/2}}$

$$\Rightarrow \quad L = \frac{c \cdot T}{2}\left(1 - v^2/c^2\right) = \frac{c\,T'}{2}\sqrt{1 - v^2/c^2} =$$

$$= L'\sqrt{1 - v^2/c^2}$$

contraction $\quad \boxed{L = L_0\sqrt{1 - v^2/c^2}}$

observed length is shorter !

moving object squeezed in prop. direction

relativistic ball ...



$v \approx c\,\frac{4}{5} \longrightarrow \quad \sqrt{1 - \frac{v^2}{c^2}} = \frac{3}{5}$

# Relativity 2

## Lorentz transformation

S and S' two inertial systems

S (I) observe something at x and t

Q: where and when will this occur for S'

- linear motion in S → also linear in S'

$$\Rightarrow \quad x' = \alpha x + \beta t$$
$$t' = \gamma x + \delta \cdot t$$

- origin of S': $x' = 0$ follows straight line $x = v \cdot t$

$$\Rightarrow \quad x' = G \cdot (x - v \cdot t)$$

principle of relativity:

$$x = G(x' + v t')$$

light emitted at $x = t = 0 \Leftrightarrow x' = t' = 0$

absorbed at x at t (S)

x' at t' (S')

$$\frac{x}{t} \overset{!!}{=} \frac{x'}{t'} = c \quad \Rightarrow \quad x = G(c + v) t'$$

$$x' = G x \left(1 - \frac{v}{c}\right)$$

$$c = \frac{x'}{t'} = c^2 \left(1 - \frac{v^2}{c^2}\right) \cdot c$$

$$G = \frac{1}{\sqrt{1 - v^2/c^2}} \equiv \gamma \quad \Rightarrow \quad x' = \frac{x - v \cdot t}{\sqrt{1 - v^2/c^2}}$$

$$x = \frac{x' + v t'}{\sqrt{1 - v^2/c^2}} \Rightarrow \ldots \Rightarrow t' = \frac{t - \frac{v \cdot x}{c^2}}{\sqrt{1 - v^2/c^2}}$$

## Lorentz - transformation

$$x' = \frac{x - v \cdot t}{\sqrt{1 - v^2/c^2}} \qquad t' = \frac{t - \frac{v \cdot x}{c^2}}{\sqrt{1 - v^2/c^2}}$$

Inverse:

$$x = \frac{x' + v t'}{\sqrt{1 - v^2/c^2}} \qquad t = \frac{t' + \frac{v \cdot x'}{c^2}}{\sqrt{1 - v^2/c^2}}$$

Remark: introduce space-time coordinates

$$(ct, x) \equiv (x^0, x^1)$$

$$\begin{pmatrix} c \cdot t' \\ x' \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1 - v^2/c^2}} & \frac{-v/c}{\sqrt{1 - v^2/c^2}} \\ \frac{-v/c}{\sqrt{1 - v^2/c^2}} & \frac{1}{\sqrt{1 - v^2/c^2}} \end{pmatrix} \begin{pmatrix} c \cdot t \\ x \end{pmatrix}$$

include also $y$ and $z$ $\Rightarrow$

four-vector: $\{x^\mu\} \equiv X = (ct, x, y, z) = (ct, \underline{x})$

$\mu = 0, 1, 2, 3$

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{1 - v^2/c^2}} & \frac{-v/c}{\sqrt{1 - v^2/c^2}} & & \\ \frac{-v/c}{\sqrt{1 - v^2/c^2}} & \frac{1}{\sqrt{1 - v^2/c^2}} & & \\ & & 1 & \\ & & & 1 \end{pmatrix}}_{\Lambda = \{\Lambda^\mu{}_\nu\}} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

- $(x^\mu)' = \sum_\nu \Lambda^\mu{}_\nu \, x^\nu$ or $X' = \Lambda X$

- $\det \Lambda = 1$

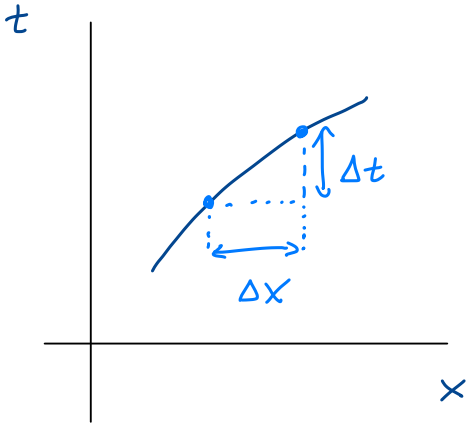
easy to check:

$$(\text{"length"})^2 \equiv c^2t^2 - x^2 - y^2 - z^2 = c^2t'^2 - x'^2 - y'^2 - z'^2$$

↑ hyperbolic length of a four-vector is independent of observer!

Proper time:

bug flies in x-direction:




$$u = \frac{\Delta x}{\Delta t}$$

For another observer

$$\begin{pmatrix} c\Delta t' \\ \Delta x' \end{pmatrix} = \frac{1}{\sqrt{1 - v^2/c^2}} \begin{pmatrix} 1 & -v/c \\ -v/c & 1 \end{pmatrix} \begin{pmatrix} c\Delta t \\ \Delta x \end{pmatrix}$$

$$\Rightarrow \quad c^2\Delta t'^2 - \Delta x'^2 = c^2\Delta t^2 - \Delta x^2$$

(check!)

for the bug: $x' = 0 \quad \Delta x' = 0$ !

$$c^2 \Delta\tau^2 = c^2\Delta t^2 - \Delta x^2 = \Delta t^2 \left(c^2 - \underbrace{\frac{\Delta x^2}{\Delta t^2}}_{v^2}\right)$$

↑ proper time

$$\Rightarrow \quad \Delta\tau^2 = \Delta t^2 \left(1 - v^2/c^2\right) \quad \Rightarrow \quad \Delta\tau = \Delta t \cdot \sqrt{1 - v^2/c^2}$$

$$\Rightarrow \quad \boxed{\Delta t = \frac{\Delta\tau}{\sqrt{1 - v^2/c^2}}}$$

time dilation!

# Velocity - addition

S: observes changes : $dx$ , $dt$

S': $dx' = \Gamma (dx - v\,dt)$

$dt' = \Gamma \left(dt - \frac{v}{c^2} dx\right)$

$$\Downarrow$$

$$u_x' = \frac{u_x - v}{1 - \frac{v}{c^2} \cdot u_x}$$

similarly $u_x = \dfrac{u_x' + v}{1 + \frac{v}{c^2} u_x'}$

- $u_x' \approx 0 \Rightarrow u_x = u_x' + v$ ✓ (Galileo)
- $u_x' = c \Rightarrow u_x = c$ !

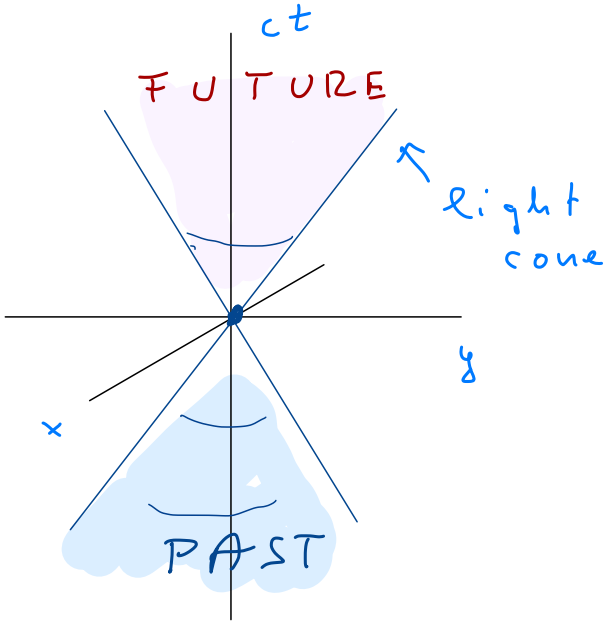
Nothing can propagate faster than light!

## Minkowski space:

space - time

space of four-vectors $x^\mu$

emit light at $x = 0$, $t = 0$ in all directions:




$\sqrt{x^2 + y^2 + z^2} = ct$

"cone"

PAST can influence me

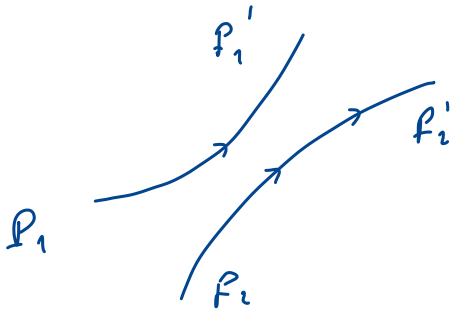
I can influence FUTURE!

# Equation of motion (Newton)

momentum conservation is crucial

(1) it is related to homogeniety of space

(2) it is the heart of Newton's laws




definition of force

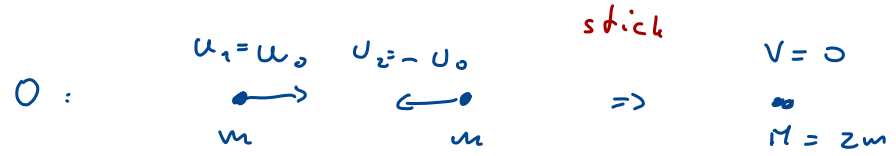
$$\frac{d\underline{p}_1}{dt} = \underline{F}_{1\leftarrow 2}$$

$$\frac{d\underline{p}_2}{dt} = \underline{F}_{2\leftarrow 1}$$

Newton $\underline{F}_{1\leftarrow 2} = -\underline{F}_{2\leftarrow 1}$

$$\Rightarrow \quad \frac{d}{dt}(p_1 + p_2) = 0 \quad \Rightarrow \quad p_1 + p_2 = p_1' + p_2'$$

problem with momentum conservation




O': $v \equiv u_0$ (co-moving with body 1)

A) velocity transformation: $v' = \dfrac{v - u_0}{1 - \frac{v \cdot u_0}{c^2}} = -u_0$

B) momentum conservation:




$$m u_1' + m u_2' = 2m \cdot V'$$

$$\Rightarrow \quad V' \overset{??}{=} -\frac{u_0}{1 + u_0^2/c^2} \quad \text{contradicts velocity transformation}$$

solution: mass depends on velocity

$$m = \frac{m_0}{\sqrt{1 - u^2/c^2}} \qquad m_0 \text{ rest mass}$$

claim: momentum conservation is then OK

( Remark: $M_0 \neq 2m_0$ in this case, but

$$M_0 = \frac{2m_0}{\sqrt{1 - v_0^2/c^2}}$$ , see later )

rel. momentum:

$$\underline{p} = \frac{m_0 \underline{u}}{\sqrt{1 - \underline{u}^2/c^2}}$$

Newton's equation:

$$\underline{F} = \frac{d\underline{p}}{dt} \; ; \qquad \underline{F} = e\underline{E} + e\,\underline{u} \times \underline{B}$$

$$\underline{E} = \underline{E}(\underline{x}, t)$$

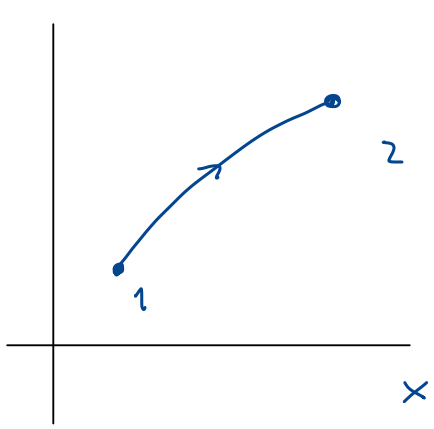
$$\underline{B} = \underline{B}(\underline{x}, t)$$

remark: $\underline{p} = m_0 \frac{d\underline{x}}{d\tau}$ ← proper time $\neq m_0 \frac{d\underline{x}}{dt} = m_0 \cdot \underline{v}$

$$\frac{d\underline{x}}{d\tau} = \frac{dt}{d\tau} \cdot \frac{d\underline{x}}{dt} = \frac{1}{\sqrt{1 - \underline{v}^2/c^2}} \cdot \underline{u} \qquad \checkmark$$

## Energy:

Work of force on particle




( drop $y, z$ for now )

$$W = \int_{x_1}^{x_2} dx\, F = \int_1^2 dx\, \frac{dp}{dt} = \int_1^2 dt\, u \frac{d}{dt}\left(\frac{m_0 u}{(1-v^2/c^2)^{1/2}}\right) =$$

$$\frac{d}{dt}\left(\frac{m_0 u^2}{(1-v^2/c^2)^{1/2}}\right) - \frac{m_0 u \dot{v}}{(1-v^2/c^2)^{1/2}}$$

$$= \left(\frac{m_0 v^2}{\sqrt{\phantom{x}}}\right)_1^2 - \int_1^2 dv\, \frac{m_0 v}{(1-v^2/c^2)^{1/2}} = \left(\frac{m_0 v^2}{\sqrt{\phantom{x}}} + m_0 c^2 \sqrt{\phantom{x}}\right)_1^2 =$$

$$- m_0 c^2 \sqrt{1 - v^2/c^2} \qquad \frac{m_0 c^2}{\sqrt{\phantom{x}}}\left(1 - \frac{v^2}{c^2}\right)$$

$$= \left(\frac{m_0 c^2}{\sqrt{\phantom{x}}}\right)_1^2 = \frac{m_0 c^2}{\sqrt{1 - v_2^2/c^2}} - \frac{m_0 c^2}{\sqrt{1 - v_1^2/c^2}} \equiv E_2 - E_1$$

$W = E_2 - E_1$ with

$$E \equiv \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} = m \cdot c^2$$

energy

relativistic mass

For small $v$ :

$$E = m_0 \cdot c^2 + \frac{m_0}{2} u^2 + \dots$$

rest energy

kinetic energy

## Four - momentum

$$\underline{p} = m_0 \frac{d\underline{x}}{d\tau} \qquad p^0 = ?$$

$$p^0 = m_0 \frac{dx^0}{d\tau} = m_0 \frac{c\,dt}{d\tau} = \frac{m_0 c}{(1 - \underline{v}^2/c^2)^{1/2}} \overset{!}{=} \frac{E}{c}$$

four - momentum !

$$P = \left(\frac{E}{c}, \underline{p}\right) = \left(\frac{m_0 c}{\sqrt{\quad}}, \frac{m_0 \underline{u}}{\sqrt{\quad}}\right)$$

- all components conserved in closed systems

$\mu = 0 \leftrightarrow$ energy conservation

$\mu = 1,2,3 \leftrightarrow$ momentum conservation

energy $\leftrightarrow$ time

momentum $\leftrightarrow$ space

Lorentz - transformation mixes energy and momentum:

$$P' = \Lambda \cdot P$$

- it's "length" is the same for anyone

$$P^2 = \frac{m_0^2 c^2}{1 - v^2/c^2} - \frac{m_0^2 u^2}{1 - v^2/c^2} = m_0^2 c^2 = \text{cst} !$$

"length" of four-momentum

$$\left(\frac{E}{c}\right)^2 - p^2 = \frac{m_0^2 c^2}{1-u^2/c^2} - \frac{m_0^2 u^2}{1-u^2/c^2} = m_0^2 c^2$$

$$\Rightarrow \quad \boxed{E = \sqrt{m_0^2 c^4 + p^2 c^2}}$$

energy-momentum relation

- if $|p| \gg m_0 c \quad \Rightarrow \quad E \approx |p| \cdot c$
- for light (photons) $m_0 = 0 \Rightarrow E = |p| \cdot c$

$$E = \frac{m_0 c^2}{\sqrt{\quad}}, \quad p = \frac{m_0 u}{\sqrt{\quad}} \Rightarrow u = \frac{p \cdot c^2}{E} = \frac{p c^2}{(p^2 c^2 + m_0^2 c^4)^{\frac{1}{2}}}$$

$$\boxed{u = \frac{\partial E}{\partial p}} \qquad \text{if } E = |p| \cdot c \Rightarrow u = c \text{ !}$$

massless particles propagate with speed of light

Energy conservation has curious consequences




$$E_{in} = 2 \cdot \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}}$$

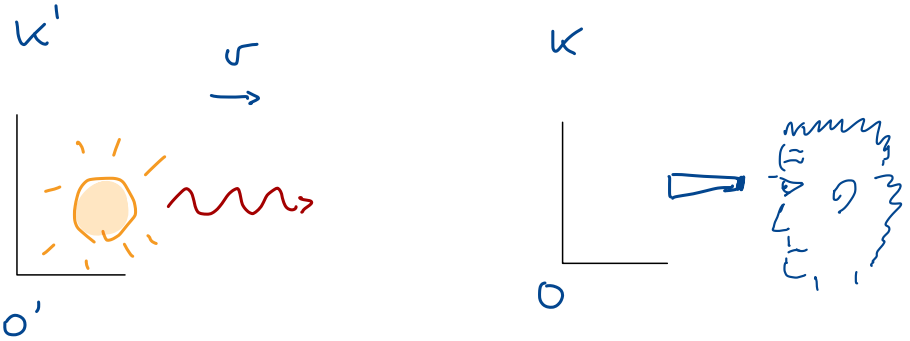
$$E_f = M \cdot c^2 = E_{in}$$

$$M = \frac{2 m_0}{\sqrt{1 - v^2/c^2}} > 2 m_0 \; !$$

internal energy increases mass !!!
(binding energy decreases mass)

## Doppler effect:

Q: star emits light with frequency $f' = f_0 = \frac{1}{T_0}$
=> What $f$ do we measure




- observer at star:
  emission of one wave front in time

$$\Delta t' = T_0 = \frac{1}{f_0}$$

- we observe a wave length in time $\Delta t = \frac{T_0}{\sqrt{1 - v^2/c^2}}$

$$\lambda = c \Delta t - v \Delta t = (c - v) \cdot \Delta t = c \cdot T_0 \left( \frac{1 - v/c}{1 + v/c} \right)^{1/2}$$

$$\Rightarrow \quad f = \frac{c}{\lambda} = f_0 \left( \frac{1 + v/c}{1 - v/c} \right)^{1/2}$$

relativistic Doppler shift

Applications:

- astronomy (red shift)
- Radar
  measure motion of targets
  air
- Laser cooling → quantum technology

(1997) Chu, Cohen-Tannoudji, Phillips

$\Rightarrow \quad T \approx 10^{-10} - 10^{-12}\ K$

# Energy and mass

in atomic/nuclear/particle physics we measure energy in $e \cdot V$ :

$$1\ eV = 1{,}60 \cdot 10^{-19}\ C \cdot 1V = 1{,}6 \cdot 10^{-19}\ J$$

↑ energy got by charged particle accelerated through 1 V

We can measure mass also in eV :

$$m_0 \rightarrow m_0 c^2$$

Electron mass, e.g. :

$$m_0 \cdot c^2 = 9{,}11 \cdot 10^{-31}\ kg \cdot \left(3 \cdot 10^8 \tfrac{m}{s}\right)^2 =$$

$$= 82 \cdot 10^{-15}\ J = \frac{8{,}2 \cdot 10^{-14}}{1{,}6 \cdot 10^{-19}}\ eV = 5.12 \cdot 10^5\ eV$$

$$= 511\ keV = 0{,}511\ MeV$$

Q: Accelerate electron by 500 kV → what will be u/c ?

A:

$$E = m_0 c^2 + eV = 1011\ keV = \frac{511\ keV}{\sqrt{1 - v^2/c^2}}$$

$$1 - \frac{v^2}{c^2} = \left(\frac{511}{1011}\right)^2 = 0{,}255$$

$$\frac{v^2}{c^2} = 0{,}744 \Rightarrow \underline{\underline{v = 0{,}86 \cdot c}}$$

# Binding energy and fusion

proton

$M_p c^2 = 938{,}27$ MeV

neutron

$M_n c^2 = 939{,}57$ MeV

$M_n c^2 - M_p c^2 = 1{,}3$ MeV $> m_e c^2$

n is unstable:

$n \rightarrow p + e^- + \bar{\nu}_e$ ($\sim 15$ min)

↑ massless, chargeless spin 1/2 particles

## Deuteron:

$M_D \cdot c^2 = 1875{,}61$ MeV $= M_p c^2 + M_n c^2 - 2{,}23$ MeV

$M_p c^2 + M_n c^2 = 1877{,}84$ MeV

2,23 MeV: binding energy $E_b$

n decay is not possible!

nuclear force stabilizes n

$$M_{bound} c^2 + E_b = \sum_i m_i c^2$$

↑ constituents

Deuteron creation in Sun:

$p + p \rightarrow {}^2H + e^+ + \nu$

## fusion

$D + T \rightarrow {}^4He + n + 17{,}6$ MeV

${}^2H$ ${}^3H$

↑ binding energy is released

(chemical reaction → 1 - 10 eV

→ $10^6$ times more energy / reaction)